

Iowa Renewable Energy GHG Inventory:

Comparing Electricity Generation Sources for a Greener Future

Emma Pingel

Department of Ecosystem Science & Sustainability

ESS 545: GHG Inventories

May 12, 2024

Executive Summary

While renewable energy is more efficient and known to be eco-friendly, emissions are still generated from the production, transportation, and maintenance of these systems. These embodied contributions to atmospheric greenhouse gases are important factors when considering the impact of transitioning to renewable sources of electricity generation like wind. Thus, the goal of the study was to compare the emissions from wind-based electricity to more traditional sources such as coal and natural gas in the state of Iowa to inform policymakers interested in diversifying the energy mix. A current inventory of emission trends for coal, natural gas, petroleum, and wind resulting in the latter energy source generating the least amount of emissions in 2023 at 974 *TMT CO₂* with uncertainty ranging from 394 *TMT CO₂* to 2029 *TMT CO₂*. After projecting three mitigation scenarios, the business-as-usual emission trend was identified to be most optimal as it was calculated to generate 23,198 *TMT CO₂* given upper and lower limits of 18,258 *TMT CO₂* and 29,655 *TMT CO₂* for 2050. While the project was not considered viable, it is recommended that Iowa continue to invest in renewable energy sources to fulfill the state's energy initiatives and increase security.

Introduction

Life is bursting with energy from the commute between locations to the process of winding down before bed. Public transit, garage doors, electronic chargers, screen projectors, household appliances, etc; all of these activities require electricity to achieve their functions. This industry has a wide array of generation sources, each with unique operating potential based on the criteria for meeting electricity demand. Renewable energy, specifically, has emerged as a “sustainable” solution with similar electricity output quantitatively and lesser environmental impact. Methods of production might include leveraging nearby water ways to generate electricity via hydropower, installing photovoltaic panels on top of homes to collect solar power, constructing turbines to rotate with the wind, and tapping into underground climates that offer a geothermal exchange of heat (Nunez n.d.). Many opportunities exist in the realm of electricity as local geographic features, economic incentives, and social norms contribute to the use of renewable energy sources. Wind, especially, has increased in popularity across the nation over the past few decades since its first onshore, utility-scale turbine farm in 1980 on California's

Altamont Pass which surpassed 100 Gigawatts of production capacity in 2019 (USDOE n.d.). According to the Office of Energy Efficiency & Renewable Energy (2015), a singular Gigawatt is the equivalent of 100 million LED lightbulbs or 2,000 Corvette Z06s.

This market shift in energy supply has been supported by the durability of wind turbines and their minimally invasive nature. In order to efficiently harness the planet's wind energy for utility-scale purposes, these structures must be around 100 meters tall so they can interact with stronger currents that move at faster rates (USEIA 2023). The turbines' long blades are then rotated around that shaft to charge the generator with mechanical energy for conversion to electrical energy. Throughout this electricity generation process, there is a negligible amount of greenhouse gas emissions generated since the turbines utilize a natural process that is not created for the sole purpose of rotating turbine blades. However, manufacturing the parts that construct a turbine, transporting them to their identified field site, and maintaining their quality to ensure long-lasting performance contribute to an embodied emission for each turbine. This is especially relevant for states that have turned to wind energy as an alternative electricity generation method to power its residents, businesses, and everything in between.

Iowa is a prime example of the shift towards wind energy as its topography in the middle of the great plains exposes this region to consistent wind patterns that increase in speed as they blow through the open landscape. This suggests a large potential for renewable energy investment in wind turbines because of the almost guaranteed productivity year-round, which is reflected in Figure 1's sudden intensification of wind energy. The state's energy portfolio expansion is in response to Iowa's goal of simultaneously boosting economic growth, improving energy security, and maintaining environmental integrity (IOEI 2007). Harnessing wind energy allows Iowa to continue promoting a healthy economy as approximately 60.5 million USD was collected in 2022 from tax revenue, especially since the state believes turbines can "be integrated at high levels without significant issue" (IEC 2023). Regardless, embodied emissions from wind turbines challenge the environmental quality of their monetary and accessibility benefits. It is important to consider these impacts as most wind turbines are not constructed on site and, despite the handful of manufacturing facilities present in eastern Iowa (Wind Exchange n.d.), will need to be driven to the west side of the state via semi-truck where most wind farms are located. Thus,

greenhouse gas emissions from wind energy need to be estimated for comparison with other, nonrenewable sources of energy.

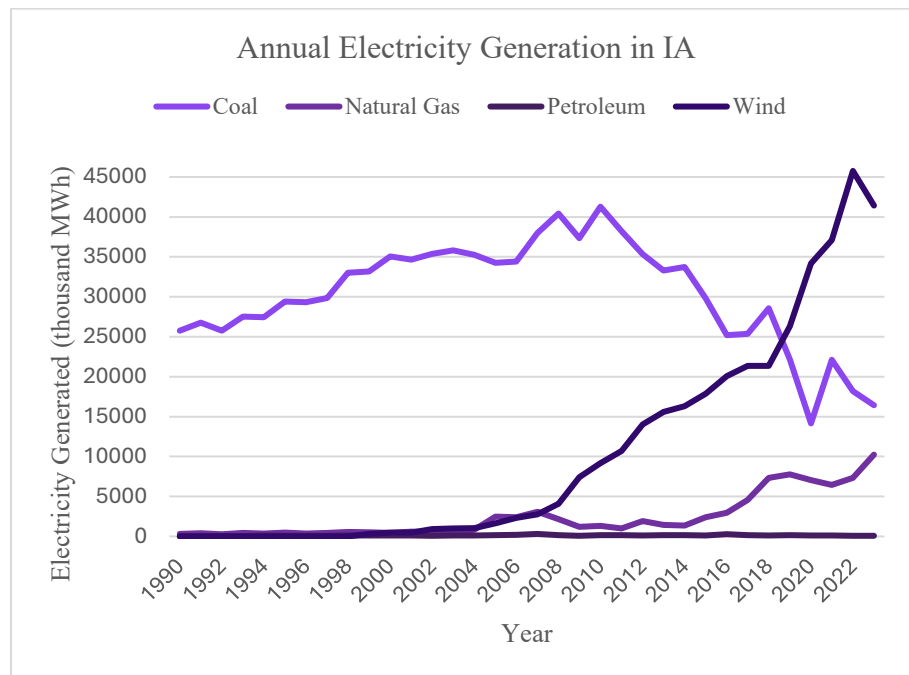


Figure 1. Annual electricity generation trends since 1990 in the state of Iowa, specifically focusing on coal, natural gas, petroleum, and wind as energy sources.

Iowa's current energy portfolio is mostly composed of coal, natural gas, petroleum, and wind with a significant increase in wind reliance since the early 2000s. As previously mentioned, this energy source produces embodied emissions from the manufacturing, transportation, and maintenance processes; therefore, a positive correlation can be assumed between emissions and production. Estimates of these contributions will be based on existing literature and reported in grams of carbon dioxide equivalents ($g CO_2$) as emission factors. The objective of this study is to compare embodied wind emissions to the other three electricity production sources in the state of Iowa to determine if this alternative is the best method for reducing energy emissions while also exploring how these systems can be effectively implemented.

Methods

Emission estimates produced in the study comply with Intergovernmental Panel on Climate Change (IPCC) 2006 Guidelines which highlight a general equation for emissions as seen below.

$$Emissions = AD \times EF$$

Where:

AD = activity data describing the human contribution to emissions eg. electricity production

EF = emission factor quantifying emissions or removal per unit activity

Activity data is collected in megawatt hours (MWh) using a top-down approach to quantify the total amount of electricity produced per energy source in Iowa each year. When estimating emissions from electricity production in Iowa, the study adopted the methods in IPCC's 2006 Guidelines for Approach II with a Monte Carlo simulation approach (IPCC 2006 1.3) for each energy source's emission factors. This approach has a fairly simple procedure that outlines the specification of categorical uncertainties, selection of a random value to start iterations, estimation of the annual emissions, and storage of the results. Such analyses are necessary to quantify any random errors or bias in the estimate. Thus, Approach II was applied to estimate electricity production emissions in Iowa with 10,000 iterations of a Monte Carlo simulation and a 95% confidence interval for each nonrenewable energy source in R Version 4.4.3.

As an integral industry to Iowa's functions, the US Energy Information Administration has recorded annual electricity production data in MWh starting in the year 1990 and going through 2023 for coal, natural gas, petroleum, wind, nuclear, solar, hydroelectric, and biomass sources (USEIAa 2024, Table 5). This State Energy Profile included source categories that were irrelevant to the study because of their nominal contribution to Iowa's energy portfolio so nuclear, solar, hydroelectric, and biomass energy sources were not analyzed. Using IPCC Tier 2 methods for calculating emissions factors from external sources, the CO_2 for the three nonrenewable sources in this study were referenced from a life cycle assessment originally conducted by the National Renewable Energy Laboratory (NREL) in 2013. In a 2021 update,

NREL estimated median greenhouse gas emission values from one-time upstream activities like material acquisition, ongoing combustion where applicable, ongoing non-combustion from operation and maintenance, and one-time downstream emissions after plant decommissioning. The resulting emissions factors are as follows: 1001 $g\ CO_2/kWh$ for coal, 486 $g\ CO_2/kWh$ for natural gas, 840 $g\ CO_2/kWh$ for petroleum, and 13 $g\ CO_2/kWh$ for wind electricity production (Table 1). This justification followed the same framework as the embodied emission factors for wind and therefore were more relevant to compare annual electricity production emissions. These values were selected over IPCC 2006 emission factors because they are more cohesive with the embodied emissions of wind.

Table 1. Energy sources for electricity production in Iowa and their emission factor estimates in $g\ CO_2/kWh$.

Source	Preferred	Minimum	Maximum
Coal	1001	675	1689
Natural Gas	486	410	650
Petroleum	840	510	1170
Wind	13	7	56

Additionally, USEIA data is not reported with any uncertainty ranges because the organization sources energy data directly from the power plants producing electricity. It is improbable that this is an accurate representation of emissions because some of the input source or electricity output can be assumed lost throughout the manufacturing, transportation, or maintenance processes. Thus, the study inferred a $\pm 3\%$ level of uncertainty to calculate electricity production standard deviation and complete the 95% confidence interval around each energy source. To account for the variation in selected energy source emission factors, a triangular probability density (Figure 2) was simulated with upper and lower limits recognized by IPCC's AR5 Working Group III in Chapter 7 (2014). Similarly, a normal probability density function (PDF) was used to simulate production amounts since uncertainty was not provided by USEIA. All of NREL's 2021 median estimates are within IPCC's AR5 emission boundaries, verifying the values for application in the study. Only CO_2 emissions are included in the conservative carbon inventory because they are the main contributors to greenhouse gas emissions for electricity production. While methane and nitrous oxide emissions are relevant to

electricity production activity, the benefit of CO_2 removals are most appropriate for the inventory since CO_2 is most abundant.

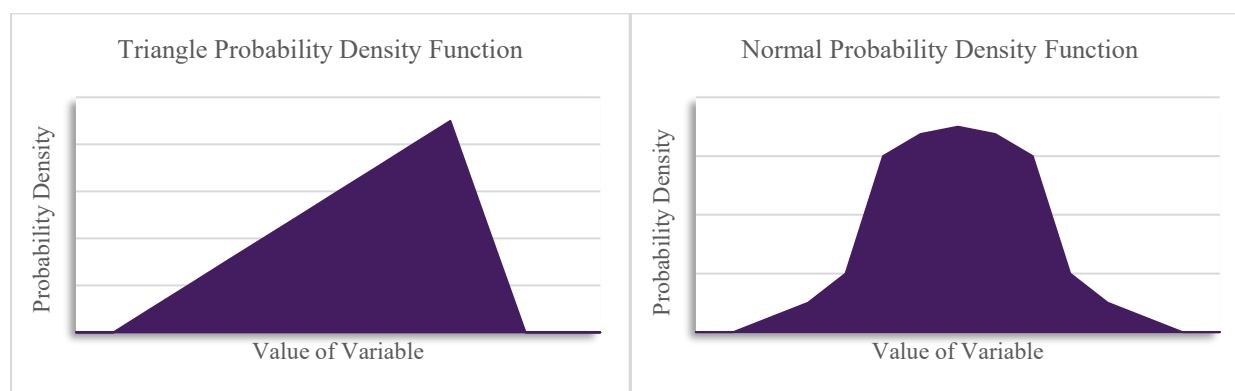


Figure 2. Triangle PDF describing the known range of values and preferred value for determining emission factors in an uncertainty analysis (left). Normal PDF illustrating the assumed uncertainty in production amounts (right).

Given the current electricity production trends in the state of Iowa, a historical scenario mirrors the diversifying portfolio of energy sources between 1990 and 2023. Looking towards the future, a business-as-usual (BAU) scenario is drafted based on the existing infrastructure for electricity production utilizing mostly coal, natural gas, and wind. The first mitigation scenario (M1) examines the increasing investment in natural gas electricity production seen in Figure 1 with a 45% amplification of this energy source by 2030 and another 40% increase by 2050. A statewide initiative has already been set in motion with Iowa's largest energy suppliers, MidAmerican Energy and Alliant Energy, committing to the construction of a natural gas power plant in Adair County (Eller 2015) so M1 is comparable to the state's current activities. The second mitigation scenario (M2) predicts how Iowa's energy portfolio and emissions would be impacted by a comprehensive transition to wind-based electricity production. Chinmoy and their colleagues (2019) studied the social and economic benefit of a scenario like M2, highlighting the enhanced quality of living for residents and improved reliability of wind power. Coal power plants would be eliminated from the energy portfolio with production goals reallocated to wind turbines. BAU, M1, and M2 would all project emissions through 2050 for all Iowans as the state energy goals previously identified in the study.

The study has potential to experience leakage in emissions as large-scale electricity production from wind might prompt consumers to use more as a cobra effect. This could lead to

more emissions that are not considered due to the project's boundary. Wind is a biological process that will exist with or without turbines, so permanence is not foreseen as an issue because there is little uncertainty in the viability of this energy source. Internal risks may present themselves in the form of opportunity costs during implementation, but the price of wind energy should decrease like they already have across the nation since 2007 (Bolinger & Wiser 2014). The most probable source of risk comes from external entities such as policymakers and land ownership, especially with interest group influence in wildlife matters and the transboundary transmission network (USDOEa n.d.). Altogether, emission reduction credits are calculated according to the quantitative differences between each energy source and their corresponding electricity production emissions. This mitigation measure sets aside a percentage of the aggregate, verified carbon removals to buffer losses from the risks discussed above and can be adjusted at a later date with continued project success.

Assumptions

Activity data collection is crucial to an accurate and precise estimate so spending more time on research is expected. During the review of existing literature, it became evident that some of the variables had potential for variation so a few assumptions should be noted to ensure full transparency throughout the inventory. The wind turbines referenced in this study are solely focused on horizontal-axis systems at the utility-scale with average maintenance requirements and similar travel times between the manufacturing plant and final site. Likewise, coal, natural gas, and petroleum sources are generalized in terms of electricity production inputs; therefore, data reflects total output per year for that source. The study also assumes that no natural or unforeseen events occur with typical weather conditions and economic activity.

Results & Discussion

Emissions from electricity production in the state of Iowa were estimated for eight intervals starting in 1990 and ending in 2023 (Table 2). While petroleum is calculated to generate the least amount of CO_2 emissions in 2023 (53.5 *TMT CO₂* with uncertainty ranging from 37.2 *TMT CO₂* to 69.92 *TMT CO₂*), this energy source is not overly relevant to the state's

portfolio due to the minimal electricity production output (Figure 1). Hence, wind has the smallest emission output from electricity production in the same year at 974 *TMT CO₂* with uncertainty ranging from 394 *TMT CO₂* to 2029 *TMT CO₂* compared to coal's estimate around 17,987 *TMT CO₂* with uncertainty ranging from 12,623 *TMT CO₂* to 25,595 *TMT CO₂* and natural gas's 5,220 *TMT CO₂* emissions estimate with uncertainty ranging from 4,413 *TMT CO₂* to 6,340 *TMT CO₂*. Wind's drastically different emission trends over the eight intervals were expected due to the knowledge of contributions stemming from embodied activities like manufacturing, transportation, and maintenance. This corresponds to Bhandari et al. (2018) who reviewed the differences between turbine size and capacity factors, concluding that larger turbines with greater capacity had lower emissions which is reflected in the study's utility-scale, horizontal-axis turbines.

Table 2. Emission estimates for the four main energy sources in Iowa (TMT CO₂).

Year	Coal	Natural Gas	Petroleum	Wind
1990	28,205.9	169.8	45.8	0
1995	32,338.8	244.5	55.5	0.003
2000	38,401.3	220.7	85.7	11.6
2005	37,469.6	126.4	125.9	38.8
2010	45,232.4	667.9	129.9	215.9
2015	32,654.2	1,222.3	92.9	421.4
2020	15,502.6	3,585.4	93.4	803.7
2023	17,987.3	5,219.9	53.5	974.0

Another study conducted by Magazzino (2021) found a predictive causal link between coal consumption and *CO₂* emissions leading them to advocate for renewable energy use in sustainable development initiatives. Figure 1 reiterates this relationship as the higher rates of electricity production from nonrenewable energy sources in Iowa results in greater emission estimates seen in Table 2. Despite the additional increase in wind-based production, *CO₂* emissions remain below coal and natural gas throughout the study, supporting a conclusion regarding the lesser impact of such embodied emissions from manufacturing, transportation, and

maintenance. As depicted in Figure 3, the state is greatly positioned to capitalize on the sustainable alternative of wind turbine electricity to achieve its energy goals.

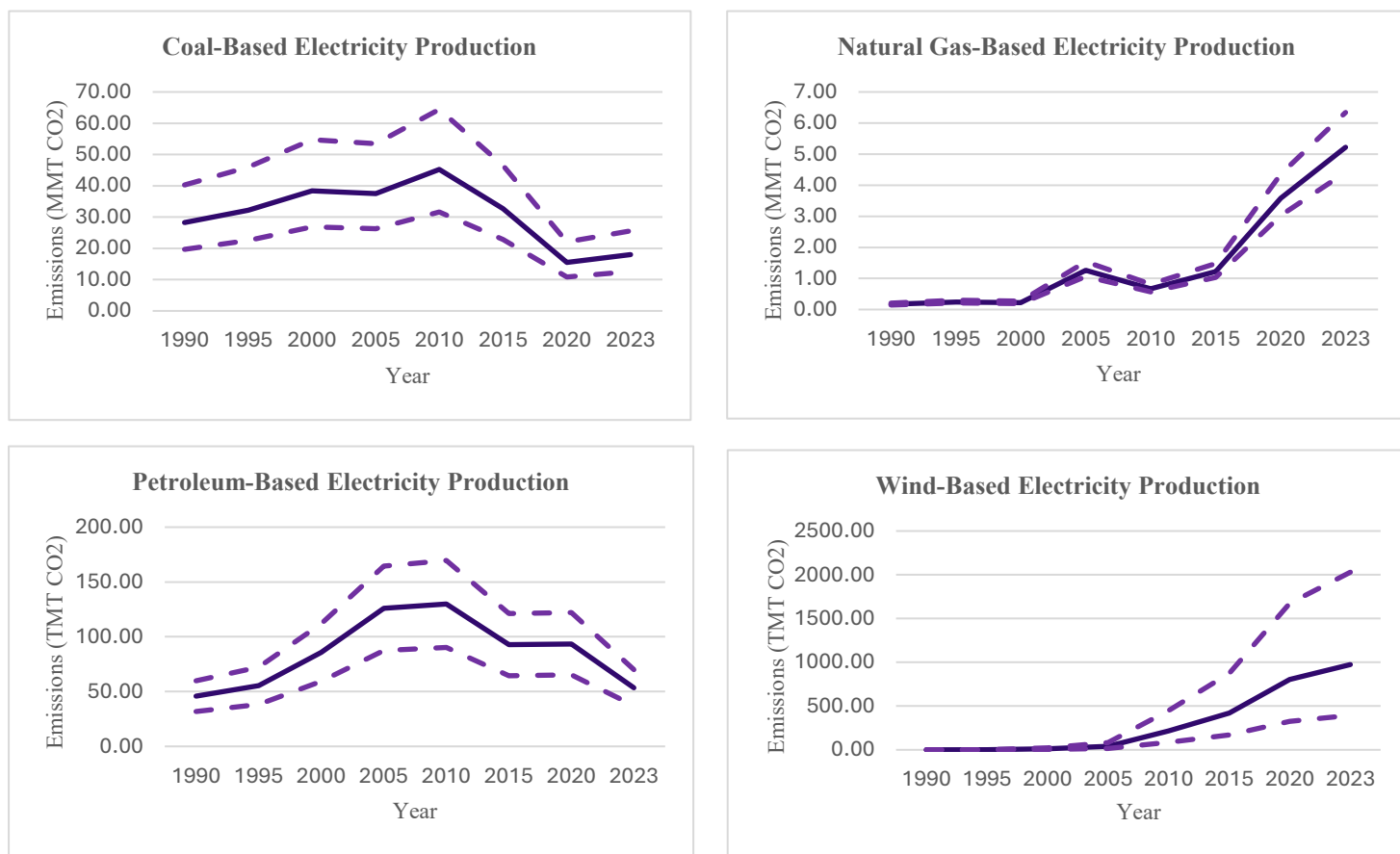


Figure 3. Estimated emissions in Iowa per energy source including uncertainty ranges.

Moreover, Iowa State University emphasized the state's mainstream sale of wind-based electricity and its potential to jumpstart the nation's energy security (McCalley and Krapfl 2021). The prime location and capacity of Iowa's wind turbines provides opportunities for more state revenue in terms of helping neighboring energy grids respond to fluctuations in use. This type of program would support Iowa's goal to identify a stable energy source for long-term independence as it encourages more investment in renewable infrastructure. However, the future scenarios outlined above should be explored to determine how emissions might change over time with varying mitigation plans.

Across all three scenarios, a 20% increase in total electricity production is expected before 2030 with an equal increment between 2031 and 2050. Following the baseline trends of

Iowa electricity production, BAU kept the same petroleum output from 2023 and reduced coal electricity production by 10% for both years of importance. The expansion of natural gas electricity production also increased wind output resulting in an estimated 23,784 *TMT CO₂* given upper and lower limits of 30,758 *TMT CO₂* and 18,542 *TMT CO₂* for 2030 and 23,198 *TMT CO₂* given upper and lower limits of 18,258 *TMT CO₂* and 29,655 *TMT CO₂* for 2050 (Table 3). Similarly, M1 amplified the change in natural gas electricity production to reflect a 45% increase from 2024 to 2030 and subsequent 40% increase from 2031 to 2050. This aggregated 25,273 *TMT CO₂* emissions given upper and lower limits of 32,345 *TMT CO₂* and 19,980 *TMT CO₂* for 2030 and 27,052 *TMT CO₂* emissions given upper and lower limits of 33,640 *TMT CO₂* and 21,913 *TMT CO₂* for 2050. Lastly, M2 focused on decommissioning all coal power plants in the state of Iowa and reallocating this demand to wind electricity production. 16,727 *TMT CO₂* emissions were estimated in this projection given upper and lower limits of 20,886 *TMT CO₂* and 13,430 *TMT CO₂* for 2030 and 23,198 *TMT CO₂* given upper and lower limits of 29,655 *TMT CO₂* and 18,258 *TMT CO₂* for 2050.

Table 3. Projected emission estimates in Iowa under BAU, M1, and M2 scenarios (TMT CO₂).

Year: Scenario	Estimate	2.5 Percentile	97.5 Percentile
2030: BAU	23,784	18,542	30,758
2050: BAU	23,198	18,258	29,655
2030: M1	25,273	19,980	32,345
2050: M1	27,052	21,913	33,640
2030: M2	16,727	13,430	20,886
2050: M2	8,790	6,988	11,287

M2 experiences the most emission reductions with its elimination of all coal-based electricity production throughout the state. This is because the combustion of coal per MWh is twice as potent as natural gas and produces up to 100 times more greenhouse gases than renewable sources (UW Electrical Engineering Dept n.d.), so removing this energy source is highly beneficial to emission reductions. The study recognizes this difference in its choice of emission factors as coal is almost eight times larger than wind's preferred factor. In terms of

policy, the study recommends following BAU trends to allow state agencies more discretion relative to the implementation of wind energy systems (Figure 4). M1 conflicts with Iowa's goals as it generates more emissions than BAU while M2 does not provide enough cushion to absorb any unexpected emissions or mistakes.

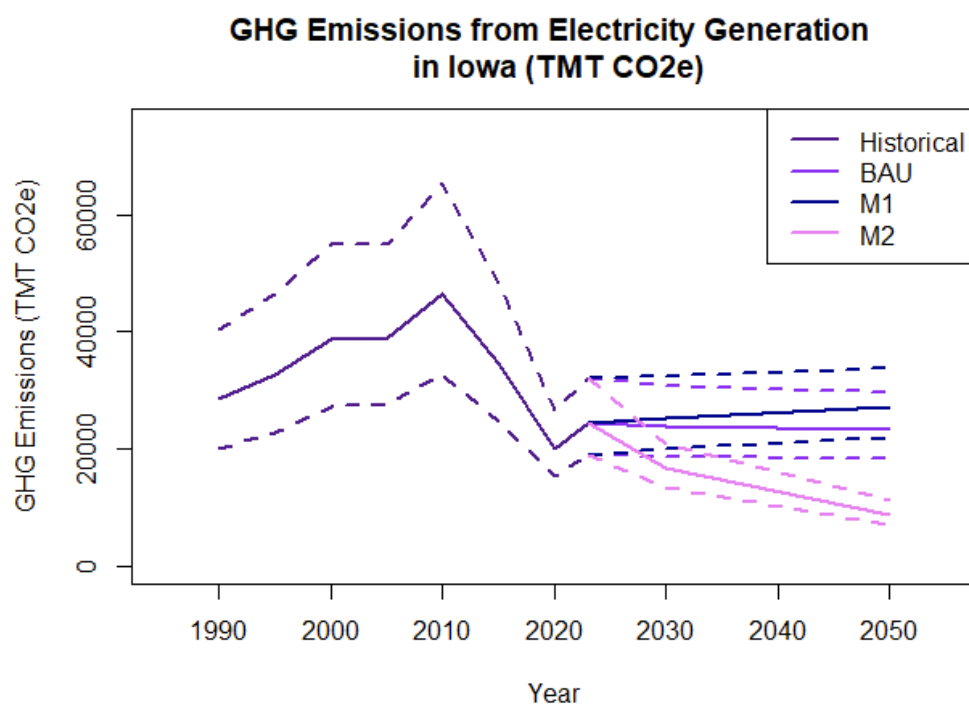


Figure 4. Mitigation scenarios for Iowa's electricity production routines relative to emissions.

In accordance with the American Carbon Registry Standards (2023), a 3-Pronged Test of Additionality also examined why emission reductions within Iowa's electricity industry might occur outside of the project's boundaries. This project failed to comply with ACR due to the existing regulatory framework enacted under the state's Renewable Portfolio Standard (RPS) in 1983 which created minimum renewable electricity generations goals for local utility companies (DSIRE 2020). Furthermore, wind turbines are largely deployed across the state, so technological barriers are minimal with institutional support for implementation in the energy portfolio. It is estimated that 59% of Iowa's total electricity output was generated by wind turbines in 2023 (USEIAb 2024), ranking the state in the top 10 for total electricity sales per capita. Production at this scale is most likely the result of a nation-wide movement to reduce CO₂ emissions which resulted in wind energy surpassing coal by 2019 (IEDA 2020). While not ideal, the study

continues to bring awareness to the issue of electricity production and greenhouse gas emissions as demand increases exponentially.

Risk analysis is also crucial to the interpretation of the study's results. Financial constraints are not foreseen with the legal endorsement from Iowa's RPS and because of the widespread adoption of wind turbines, project managers should be able to effectively oversee the execution of initiatives. Correlations between energy efficiency and a low-carbon economy have been made with the state's RPS causing many researchers to encourage greater application of the program (Ishola 2025). However, social and political influences may hinder the project from fulfilling its contractual obligation with the changing management in higher roles. Current federal administration strategies have withheld federal funding for clean energy projects and limited permit opportunities (Copley 2025). Thus, emission reduction tonnes (ERTs) were calculated according to ACR Protocol (2023) with emissions minimized to the 90% confidence interval due to the $\pm 10\%$ uncertainty requirement which the study did not satisfy and to account for any potential leakage. Since approximately two-thirds of Iowa counties have repeatedly reported high fire risks this past year because of the dry grassland climate that is vulnerable to rapid fire spread (KCCI 2025), a non-default value for natural disaster risks was used in the fire subcategory. The total ERTs at the time of credit insurance is 3,629 resulting in a buffer pool contribution of 4.5 million USD based on the Midwest Renewable Energy Tracking System's (n.d.) estimated carbon credit for the project.

The study's calculations suggest electricity production in the state of Iowa follows its business-as-usual model with expanding wind energy infrastructure and some deductions in coal dependence. While the project is not viable because of the existing policy implications and programs, the study reiterates the significant positive impact renewable energy sources have on greenhouse gas emissions from electricity production. It is recommended that Iowa continue to research and develop its energy portfolio to include more wind in order to meet the state's energy goals and build a sustainable, secure future.

References

- ACR. (2023). The ACR standard: Requirements and specifications for the quantification, monitoring, reporting, verification, and registration of project-based GHG emission reductions and removals. The American Carbon Registry. <https://acrcarbon.org/wp-content/uploads/2023/10/ACR-Standard-v8.0.pdf>
- Bhandari, R., Kumar, B. and F. Mayer. (2020). Life cycle greenhouse gas emission from wind farms in reference to turbine sizes and capacity factors. *Journal of Cleaner Production* 277. <https://doi.org/10.1016/j.jclepro.2020.123385>
- Bolinger, M. and Wiser, R. (2014). 2014 wind technologies market report. Office of Energy Efficiency & Renewable Energy, US Department of Energy. <https://docs.nrel.gov/docs/fy15osti/64522.pdf>
- Chinmoy, L., Iniyan, S., and Goic, R. (2019). Modeling wind power investments, policies and social benefits for deregulated electricity market – A review. *Applied Energy* 242. <https://doi.org/10.1016/j.apenergy.2019.03.088>
- Copley, M. (March 2025). America's clean-energy industry is growing despite Trump's attacks. At least for now. Morning Edition, National Public Radio. <https://www.npr.org/2025/03/12/nx-s1-5319056/trump-clean-energy-electricity-climate-change>
- DSIRE. (October 2020). Iowa Programs. [Database of State Incentives for Renewables & Efficiency](https://programs.dsireusa.org/system/program/ia). <https://programs.dsireusa.org/system/program/ia>
- Eller, D. (July 2015). Iowa power plants cut carbon pollution. The Des Moines Register. <https://www.desmoinesregister.com/story/money/agriculture/2015/07/14/carbon-emissions-pollution-midamerican-alliant/30162423/>
- IEC. (2023). Iowa Wind Energy Fact Sheet. Iowa Environmental Council. <https://www.iaenvironment.org/webres/File/Wind%20Energy%20Fact%20Sheet%20-%202023.pdf>
- IEDA. (2020). 2020 State Energy Report. Iowa Economic Development Authority, State Energy Office. <https://www.legis.iowa.gov/docs/publications/DF/1211198.pdf>
- IOEI. (2007). Iowa Plan for Energy Independence. Iowa Office of Energy Independence https://publications.iowa.gov/15255/1/Final_Plan.pdf
- IPCC. (2006). 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Eggleston HS, Buendia L, Mia K, Ngara T, Tanabe K, editors. Intergovernmental Panel on Climate Change, Hayama, Kanagawa, Japan, 2006.
- IPCC. (2014). Climate Change 2014: Synthesis Report. Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change [Core Writing Team, R.K. Pachauri and L.A. Meyer (eds.)]. Intergovernmental Panel on Climate Change, Geneva, Switzerland, 151 pp.
- Ishola, A. (2025). Renewable portfolio standards, energy efficiency and air quality in an energy transitioning economy: The case of Iowa. *Green Technologies and Sustainability* 3. <https://doi.org/10.1016/j.grets.2024.100159>
- KCCI. (March 2025). Iowa fire danger: Burn bans in effect for 15 counties. KCCI 8 News. <https://www.kcci.com/article/iowa-burn-bans-fire-danger-dry-weather/63632446>

- LSA. (2024). Iowa Wind Energy - 2023. Legislative Services Agency.
<https://www.legis.iowa.gov/docs/publications/FCTA/1460393.pdf>
- Magazzino, C., Mele, M., and Schneider, N. (2021). A machine learning approach on the relationship among solar and wind energy production, coal consumption, GDP, and CO2 emissions. *Renewable Energy* 167. <https://doi.org/10.1016/j.renene.2020.11.050>
- McCalley, J. and M. Krapfl. (November 2021). Macrogrid study: Big value in connecting America's eastern and western power grids. Iowa State University.
<https://www.news.iastate.edu/news/macrogrid-study-big-value-connecting-americas-eastern-and-western-power-grids>
- MRETS. (n.d.). Subscription & pricing: Choosing the right plan. Midwest Renewable Energy Tracking System. <https://www.mrets.org/subscription-costs/>
- NREL. (2021). Life cycle greenhouse gas emissions from electricity generation: Update. National Renewable Energy Laboratory, FS-6A50-80580. <https://www.nrel.gov/docs/fy21osti/80580.pdf>
- Nunez, C. (n.d.) Renewable energy explained. National Geographic.
<https://education.nationalgeographic.org/resource/renewable-energy-explained/>
- OEERE. (2015) How much power is 1 gigawatt? Office of Energy Efficiency & Renewable Energy, US Department of Energy. <https://www.energy.gov/eere/articles/how-much-power-1-gigawatt>
- UTI Blog. (2025, March 5). How much energy does a wind turbine produce. Universal Technical Institute. <https://www.uti.edu/blog/wind-turbine/how-much-energy-does-a-wind-turbine-produce>
- USDOEa. (n.d.). Advantages and Challenges of Wind Energy. Wind Energy Technologies Office, US Department of Energy. <https://www.energy.gov/eere/wind/advantages-and-challenges-wind-energy>
- USDOEb. (n.d.). History of U.S. Wind Energy. Wind Energy Technologies Office, US Department of Energy. <https://www.energy.gov/eere/wind/history-us-wind-energy>
- USEIAa. (November 2024). Iowa Electricity Profile 2023 [Table 5]. US Energy Information Administration. <https://www.eia.gov/electricity/state/iowa/>
- USEIA. (October 2023). Electricity explained. US Energy Information Administration.
<https://www.eia.gov/energyexplained/electricity/how-electricity-is-generated.php>
- USEIAb. (September 2024). Iowa State Energy Profile. US Energy Information Administration.
<https://www.eia.gov/state/print.php?sid=IA>
- UW Electrical Engineering Dept. (n.d.). The Production of Electricity Power from Coal [PowerPoint slides]. Environmental Impacts of Technology, University of Washington.
https://labs.ece.uw.edu/community/resources/ElectricityProduction_Coal.pdf
- Wind Exchange. (n.d.). Wind Energy in Iowa. Office of Energy Efficiency & Renewable Energy, US Department of Energy. <https://windexchange.energy.gov/states/ia>

Appendix

A – Quality Control Measures & Inventory Improvement Plan

The Monte Carlo simulation was evaluated by comparing the results to the deterministic result since the static value should be within the confidence interval for consistent results. Two scripts were created to (1) estimate emissions per energy source per year and (2) estimate emissions per year for mitigation purposes. Both functions resulted in the same numerical estimates and followed the same procedures in simulating emissions factors with a triangle PDF and electricity production amounts with a normal PDF. Additionally, the two scripts also used the same units to ensure resulting values could be directly compared and applied but calculations were rounded prematurely throughout the study.

Since wind emissions are assumed to be embodied in the study, confirmation was needed to verify the emission factor was accurate. Therefore, the number of wind turbines was linearly scaled back based on the belief that one turbine generates about 6,000 MWh of electricity each year (UTI Blog 2025). This calculation created a dividend around 6,910 turbines for 2023 from USEIA's total electricity production (MWh) which is supported by Iowa's reported number of turbines at 6,481 (LSA 2024). Confirming value accuracy is key to ensuring the following embodied emission calculations are reflective of plausible trends in wind development throughout the state. Technology advancements likely contributed to this industry's development at a nonlinear rate and should be researched more to improve the quality of the embodied emission estimates. Furthermore, the initial years of wind data are not available in Table 5 (USEIAa 2024) causing a time series inconsistency that should be recalculated if the study is replicated.

B – Important Links for Further Understanding

- [Monte Carlo Methodology as a part of Approach II from IPCC's 2006 Guidelines, Volume 1, Chapter 3.2.3.2](#)
- [IPCC's AR5 Working Group III in Chapter 7 on Energy Systems](#)
- [Three-Pronged Additionality Test from The ACR Standard, Version 8.0, Chapter 4.A](#)